

A Comparative Analysis of Sustained GPU Throughput and Latency Stability on Decentralized Physical Infrastructure Network(DePIN) Platform vs Centralized Cloud Platforms

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Abstract—The exponential growth of large language models has intensified demand for cost-effective GPU computing. While Decentralized Physical Infrastructure (DePIN) marketplaces like Akash promise competitive pricing, their performance characteristics relative to centralized cloud providers (Amazon Web Services(AWS), Google Cloud Platform(GCP), Azure) remain empirically unverified. This research proposes a rigorous comparative analysis of sustained training throughput, latency stability, and cost-effectiveness between Akash’s decentralized platform and major centralized providers using identical NVIDIA H100 GPU instances. Through controlled experiments involving LLaMA-2 7B fine-tuning over 8-24 hour periods, this study will measure tokens/second throughput, step-time variance, provisioning overhead and total cost-to-convergence. Expected outcomes include identification of performance bottlenecks specific to decentralized architectures, quantification of real-world cost-per-token metrics, and development of a reproducible benchmarking framework. This work addresses critical gaps by providing the first systematic, multi-hour empirical comparison of decentralized versus centralized GPU training infrastructure with standardized workloads and comprehensive telemetry.

Index Terms—Distributed training, Decentralized Physical Infrastructure Network, performance benchmarking, cost analysis

I. INTRODUCTION

A. Background

The training of modern artificial intelligence models, especially large language models (LLMs) and vision transformers, needs a lot of computing power usually sourced from centralized cloud platforms [1]. These centralized providers, like AWS, Google Cloud Platform (GCP), and Microsoft Azure, offer high-performance GPU instances backed by reliable infrastructure, but they come at high prices. This can make access difficult for academic researchers and smaller organizations [2]. Recent studies show that training deep learning models across distributed cloud resources can significantly change cost and throughput trade-offs. Using spot instances and distributed strategies can make training cheaper and more competitive with on-demand options [1].

At the same time, decentralized GPU marketplaces have come up as alternative infrastructure options, offering competitive prices through peer-to-peer resource sharing and blockchain-based coordination [3], [4]. Platforms like Akash, Vast.ai, and Golem collect unused GPU capacity from various providers, which may provide advantages over centralized cloud giants. Research on decentralized training has shown that mixed, distributed GPU clusters can achieve impressive performance gains, with speed improvements of up to 4.8 times compared to traditional centralized methods in extreme multi-city scenarios using smart scheduling algorithms [5].

However, training with distributed GPUs has its challenges. Different devices, varying network connections, and provider turnover make model-parallel and pipeline-parallel training strategies that work well in uniform data center environments more complicated [5]. Communication delays often hinder scaling efficiency, and network latency can become a serious problem in distributed settings [6], [7]. While asynchronous training methods have shown up to 20 times efficiency gains in challenging network conditions [8], the real-world, ongoing performance of decentralized marketplaces is still mostly unmeasured.

B. Research Gap

- **Sustained Multi-Hour Throughput and Variance Analysis:** Existing benchmarking studies typically report peak or short-duration throughput metrics [1], [5]. The literature lacks reproducible multi-hour training runs with comprehensive variance analysis, including jitter, thermal throttling, and host preemption specifically on decentralized marketplaces like Akash. Yuan et al. [5] study allocation using simulated geo-distributed scenarios, and FusionAI [3] describes system design with peak comparisons, but no studies provide sustained marketplace throughput variance traces under real production conditions.
- **Direct Decentralized vs. Centralized Platform Comparisons:** Extensive research exists comparing centralized cloud providers [1], and separate work explores decen-

tralized training system designs [3], [5], [8]. However, no reproducible head-to-head performance and cost studies compare decentralized marketplace platforms directly against AWS/GCP/Azure under matched ML workloads with identical hardware configurations.

- **Reproducible Cost-Per-Work Metrics:** While Erben et al. [1] report cost-throughput trade-offs for cloud spot VMs, and engineering guidance exists on cost strategies [2], the literature lacks standardized, reproducible protocols for platform-agnostic cost-per-work comparisons. Real end-to-end cost accounting including retry overhead, checkpoint transfer costs, and data egress charges are notably absent [3], [9].

C. Significance

This research will offer the first systematic comparison of decentralized and centralized GPU training systems. It will help AI and machine learning practitioners make informed choices about which infrastructure to use. By setting up reproducible benchmarking methods and measuring real-world performance against costs, this work will benefit both academic research and industry practices in deploying distributed training.

II. RESEARCH AIM

To empirically quantify and compare the sustained training performance, cost-effectiveness, and operational characteristics of decentralized GPU cloud platforms (specifically Akash) compared to the primary competitors for cloud processing like AWS, and develop reproducible metrics of comparison as well as identify platform bottlenecks and/or advantages for contemporary workloads in AI and ML.

III. RESEARCH QUESTIONS

A. RQ-Main

To what extent does the Akash DePIN platform provide a genuine, sustainable advantage in terms of performance-per-cost when compared to established industry leaders like AWS particularly for current-generation AI and machine learning workloads? My focus centers on the practicalities of fine-tuning large language models such as LLaMA-2 7B. The goal is to move beyond theoretical benchmarks and marketing claims, instead grounding the analysis in real-world training scenarios that developers and organizations encounter today.

B. RQ1 (Training Stability & Throughput)

How do distributed GPU resources on Akash compare to centralized NVIDIA H100 clusters from hyperscalers when running identical training jobs over extended periods specifically, 8 to 24 hours of continuous operation? The analysis will measure raw throughput, such as tokens processed per second, without filtering out anomalies, to provide a transparent view of performance. It's crucial to understand not just average speeds but also the consistency of those speeds: Are there periods of instability, unexpected slowdowns, or sudden spikes? Understanding the variability and predictability

of training performance will help evaluate whether Akash's decentralized approach can deliver reliability comparable to more mature, centralized solutions.

C. RQ2 (Overheads and Latency)

What are the underlying setup overheads and latencies inherent in Akash's decentralized resource provisioning? This includes the time required for container orchestration, node selection, cold starts, and any network or consensus delays that precede the commencement of actual model training. These factors will be contrasted with the rapid, often near instantaneous deployment workflows offered by centralized cloud providers via streamlined APIs. The investigation will also consider the potential for bottlenecks due to resource fragmentation, the complexity of managing distributed infrastructure, and how these issues could impact the ability to scale up or recover from failures during training jobs.

D. RQ3 (Cost-Effectiveness)

What is the comprehensive cost to completion for a full training cycle—encompassing all aspects from job initiation to convergence on Akash versus traditional cloud providers? This analysis will incorporate a granular breakdown of expenses: total spend per token processed, per epoch completed, and per successful model convergence. The evaluation will also account for all indirect costs, including those arising from failed runs, restart overheads, checkpoint migrations, data ingress and egress fees, and any additional administrative burden. The objective is to provide a transparent, apples-to-apples comparison that reflects the true economic efficiency of each platform, helping stakeholders understand where genuine savings exist and whether they justify any potential trade-offs in reliability or simplicity.

IV. RESEARCH OBJECTIVES

This study aims to establish a comprehensive and dependable benchmarking framework that enables accurate, apples-to-apples comparisons of GPU training performance across diverse cloud platforms. The fundamental objective is not simply to measure speed or throughput, but to devise and rigorously validate a standardized methodology for quantifying and reporting performance. This involves collecting a rich array of telemetry data, such as GPU utilization rates, temperature readings, and power consumption metrics to capture the full spectrum of hardware behavior under real-world training workloads. Crucially, these raw metrics are not left unanalyzed; instead, they undergo robust statistical analysis, ensuring that the findings are meaningful, reproducible, and resistant to random fluctuations or noise [1], [9].

Building upon this foundation, the investigation delves further to assess how systems perform not just in the short term, but over the extended durations that characterize modern AI training jobs. To this end, we meticulously track training throughput measured in tokens processed per second over prolonged intervals ranging from eight to twenty-four hours. This approach reveals not only average performance levels but

also uncovers underlying trends, anomalies, and degradations that may emerge over time [1], [5]. By deploying identical NVIDIA H100 80GB instances on both decentralized infrastructure like the Akash Network and on established centralized providers such as AWS we ensure that hardware variables are controlled, making performance differences attributable to the platforms themselves. Analysis of mean throughput, as well as percentile distributions (5th, 50th, and 95th), provides a nuanced understanding of both typical and edge-case behaviors, highlighting periods of consistency and volatility.

Yet, raw speed tells only part of the story. For demanding AI workloads such as the fine-tuning of large language models like LLaMA-2 7B system stability and latency become equally critical. To address this, we examine the detailed distribution of training step times, paying particular attention to tail latencies at the 99th percentile. This allows us to identify rare but significant outliers, such as spikes caused by thermal throttling, resource contention, or network jitter, which can severely disrupt training continuity. Furthermore, we dissect the “time-to-first-training-step”, the end-to-end latency from job submission through scheduling, container image pulls, and environment setup, culminating in the model’s initial forward pass [5], [8]. This breakdown exposes practical, user-facing delays, and illuminates where decentralized approaches like Akash may introduce overhead compared to the streamlined orchestration of major cloud providers.

Cost efficiency is another decisive factor, especially as large-scale training jobs can incur substantial expenses. To provide actionable insights, we perform an exhaustive cost analysis encompassing GPU hourly rates, network data transfer charges, and any costs arising from job restarts or failures. These are synthesized into intuitive, normalized metrics such as dollars spent per token processed and per epoch completed, enabling straightforward apples-to-apples financial comparisons between platforms [1], [2]. By examining these metrics in conjunction with performance data, we uncover not only which platforms deliver the best raw throughput, but also which offer the most value for money and resilience in the face of operational hiccups.

By synthesizing this wealth of telemetry, performance, stability, latency, and cost data, our study offers a holistic, evidence-based evaluation of leading cloud and decentralized platforms for AI training. This integrated analysis moves beyond headline numbers to illuminate the subtle trade-offs and operational realities that organizations must consider when selecting infrastructure for large-scale machine learning projects. Ultimately, we aim to empower decision-makers with the rigorous, transparent benchmarks required to make informed, cost-effective choices that align with their technical and business goals.

V. RESEARCH METHODOLOGY

A. Experimental Design and Workload Selection

To ensure a fair and insightful comparison, our experimental design emphasizes strict controls and comprehensive coverage across platforms [1]. All tests are conducted on identical

hardware configurations, standardizing the environment and eliminating confounding variables arising from infrastructure discrepancies. This approach allows us to isolate the performance characteristics intrinsic to each provider. Our study includes the decentralized Akash Network, which offers resource on a DePIN marketplace, as well as three leading centralized cloud providers AWS EC2 (p5.xlarge) representing the current industry standard for high-performance AI workloads.

Each workload is executed on NVIDIA H100 80GB GPU instances, selected for their state-of-the-art capabilities in large-scale AI computation. By mandating this uniform hardware baseline, we ensure that performance differences are attributable to factors such as virtualization overhead, network topology, and scheduling efficiency, rather than disparities in computational resources. Every experimental run spans 8 to 24 hours, with three to five repetitions per platform, striking a balance between capturing temporal variability and maintaining manageable operational costs [1]. This repetition not only enhances the statistical significance of our findings but also allows us to probe the consistency and reliability of each platform under sustained, production-like conditions.

Our primary workload centers on instruction fine-tuning for the LLaMA-2 7B model, utilizing widely adopted datasets such as Alpaca and Dolly-15k [10]. This choice is deliberate, fine-tuning large language models imposes intense demands on hardware, requiring substantial compute throughput, high memory bandwidth, and efficient inter-node communication. These characteristics closely mirror the requirements of real-world, high-stakes AI deployments, making our findings directly relevant to practitioners seeking to optimize for similar use cases. By focusing on a challenging and resource-intensive workload, we stress test each platform’s ability to handle the upper bounds of modern AI infrastructure demands [5], [8].

B. Data Collection and Metrics

To build a comprehensive performance profile, we adopt industry-standard benchmarking methodologies, capturing a wide range of metrics that reflect both speed and efficiency. Throughput is measured as tokens processed per second, with data aggregated into one-minute windows [1], [5]. This granularity enables us to detect transient spikes or drops in performance, providing a nuanced view that goes beyond simple averages. For each run, we calculate not only the mean throughput but also the standard deviation and key percentiles (5th, 50th, 95th, and 99th), offering insights into both typical and outlier behaviors [1]. These percentile analyses are critical for evaluating tail risk, the infrequent yet significant performance regressions that can jeopardize operational deadlines and training stability.

Latency tracking complements our throughput measurements. We monitor step duration, carefully logging any anomalies, slowdowns, or signs of thermal throttling that could indicate hardware or resource contention [5]. By maintaining a continuous record throughout each experiment, we can correlate performance dips with environmental or operational

factors, such as thermal events or background workload interference.

Our cost analysis extends well beyond headline GPU rental rates. We account for all operational expenses, including data transfer costs, storage, and any additional overhead incurred by failed runs or retries. This holistic approach yields a true cost-to-convergence metric, expressed as cost per processed token, which is indispensable for organizations seeking to maximize ROI on their AI investments [1], [2]. We further dissect the pre-training workflow, measuring elapsed time from job submission to the commencement of training. By breaking down this interval into discrete components such as scheduling latency, container image provisioning, and environment setup, we identify bottlenecks that could impede rapid iteration or scalability, providing actionable insights for platform optimization [2].

C. Data Processing and Experimental Controls

Reproducibility and rigor underpin our experimental methodology. We standardize every aspect of the software stack, deploying identical Docker containers across all platforms. The stack includes PyTorch 2.1 and CUDA 12.1, eschewing custom patches or optimizations that could introduce variability [13]. Hyperparameters are fixed, random seeds are set, and thermal baselines are established to neutralize sources of stochasticity [1], [5]. This meticulous control ensures that observed differences in performance, stability, or cost are rooted in the platform’s infrastructure, not in subtle software divergences.

Telemetry is collected at one second intervals using Prometheus, with all measurements archived in a time series database for subsequent analysis. This high-resolution monitoring allows for precise detection of both transient anomalies and persistent trends [12]. In the analysis phase, we employ descriptive statistics to summarize performance and deploy inferential techniques specifically, two-sample t-tests with a significance threshold of $\alpha = 0.05$ to rigorously assess whether observed performance differences are statistically meaningful. We further examine variance as a measure of platform stability, enabling us to compare not just average performance but also reliability over time [1]. Time-series analytical methods are applied to uncover patterns, identify recurring issues, and assess the predictability of long-running training jobs, which is critical for enterprises that require consistent service levels.

By integrating these layers of experimental rigor, data richness, and analytical depth, our study aims to provide a detailed, actionable, and trustworthy comparison of decentralized and centralized AI infrastructure options, equipping practitioners with the evidence needed to make informed deployment decisions in a rapidly evolving technological landscape.

VI. CONCLUSION

This research examines how decentralized GPU marketplaces stack up against traditional cloud providers for AI and machine learning training. It uses the same hardware,

the NVIDIA H100, and runs consistent workloads, such as LLaMA-2 7B fine-tuning and vision training. The study will conduct benchmarks lasting 8 to 24 hours to measure throughput, latency, cost, reliability, and network performance. It addresses significant gaps by providing long-duration performance data, direct comparisons between marketplaces and cloud providers, and clear cost-per-work metrics. The results will highlight real bottlenecks, hidden costs, and benefits of each platform. Additionally, it will create a reusable benchmarking framework. In the end, this research will provide the first solid evidence to help researchers and organizations make informed choices about infrastructure for large-scale AI training.

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Abstract

The exponential growth of large language models has intensified demand for cost-effective GPU computing. While Decentralized Physical Infrastructure (DePIN) marketplaces like Akash promise competitive pricing, their performance characteristics relative to centralized cloud providers (AWS, GCP, Azure) remain empirically unverified. This research proposes a rigorous comparative analysis of sustained training throughput, latency stability, and cost-effectiveness between Akash's decentralized platform and major centralized providers using identical NVIDIA H100 GPU instances. Through controlled experiments involving LLaMA-2 7B fine-tuning over 8-24 hour periods, this study will measure tokens/second throughput, step-time variance, provisioning overhead and total cost-to-convergence. Expected outcomes include identification of performance bottlenecks specific to decentralized architectures, quantification of real-world cost-per-token metrics, and development of a reproducible benchmarking framework. This work addresses critical gaps by providing the first systematic, multi-hour empirical comparison of decentralized versus centralized GPU training infrastructure with standardized workloads and comprehensive telemetry.

Keywords

Distributed training, Decentralized Physical Infrastructure Network, performance benchmarking, cost analysis, LLM fine-tuning

I. Introduction & Research Gap

Background

Modern AI model training requires substantial GPU resources from centralized clouds (AWS, GCP, Azure) at premium costs [1]. Decentralized marketplaces (Akash) promise cost advantages through P2P resource sharing [3,4], with research showing up to 4.8× speedup in geo-distributed scenarios [5].

Critical Research Gaps

- Sustained Multi-Hour Throughput and Variance Analysis:**Existing benchmarking studies typically report peak or short-duration throughput metrics [1], [5]. The literature lacks reproducible multi-hour training runs with comprehensive variance analysis, including jitter, thermal throttling, and host preemption specifically on decentralized marketplaces like Akash. Yuan et al. [5] study allocation using simulated geo-distributed scenarios, and FusionAI [3] describes system design with peak comparisons, but no studies provide sustained marketplace throughput variance traces under real production conditions.
- Direct Decentralized vs. Centralized Platform Comparisons:**Extensive research exists comparing centralized cloud providers [1], and separate work explores decentralized training system designs [3], [5], [8]. However, no reproducible head-to-head performance and cost studies compare decentralized marketplace platforms directly against AWS/GCP/Azure under matched ML workloads with identical hardware configurations.
- Reproducible Cost-Per-Work Metrics:** While Erben et al. [1] report cost-throughput trade-offs for cloud spot VMs, and engineering guidance exists on cost strategies [2], the literature lacks standardized, reproducible protocols for platform-agnostic cost-per-work comparisons. Real end-to-end cost accounting including retry overhead, checkpoint transfer costs, and data egress charges are notably absent [3], [9].

II. Research Aim

To empirically quantify and compare the sustained training performance, cost-effectiveness, and operational characteristics of Akash decentralized GPU platform versus major centralized providers for current-generation AI/ML workloads, establishing reproducible benchmarking methodologies [1,5,9].

III. Research Questions

RQ-Main: To what extent does the Akash DePIN platform provide a verifiable and sustained performance-per-cost advantage over major centralized providers (AWS, GCP, Azure) for current-generation AI/ML workloads (specifically LLaMA-2 7B fine-tuning)?

RQ1 (Training Stability & Throughput): How do the training throughput (tokens/second for LLMs) and performance stability (variance, jitter) of identically configured Akash and centralized NVIDIA H100 instances compare across 8-24 hour training runs?

RQ2 (Overheads and Latency): What are the measurable setup and initialization overheads (container provisioning time, cold start latency) introduced by Akash's decentralized resource allocation protocols compared to centralized cloud APIs?

RQ3 (Cost-Effectiveness): What is the true end-to-end cost-per-work (\$/token, \$/epoch, \$/convergence) on Akash versus centralized providers when accounting for all overhead costs including retries, checkpoint transfers, and data egress?

VI. Research Objectives

- O1: Establish reproducible benchmarking framework [1,9]
- O2: Quantify sustained throughput (8-24h runs) [1,5]
- O3: Characterize latency & stability profiles [5,8]
- O4: Measure provisioning overhead [2,9]
- O5: Calculate comprehensive cost metrics [1,2]
- O6: Identify platform-specific bottlenecks [3,5,8]

V. Research Methodology

A. Experimental Design

This study employs a controlled experimental design with matched hardware configurations across platforms to isolate infrastructure-level performance differences.



Fig. 1. Network and Hardware list for the experimental environment

B.Workload Selection(LLM Fine Tuning)

- Model:** LLaMA-2 7B [10]
- Task:** Instruction fine-tuning (Alpaca/Dolly-15k)
- Framework:** PyTorch + HuggingFace [13]

C. Data Collection

Throughput: tokens/sec [1,5]

Latency: Step time, 99th percentile [5]

Cost: \$/GPU-hr, \$/token, \$/epoch[1,2]

Provisioning: Cold start, container time[2]

Reliability:Preemptions, failures [1,2]

Host Telemetry:GPU util, temp, power[12]

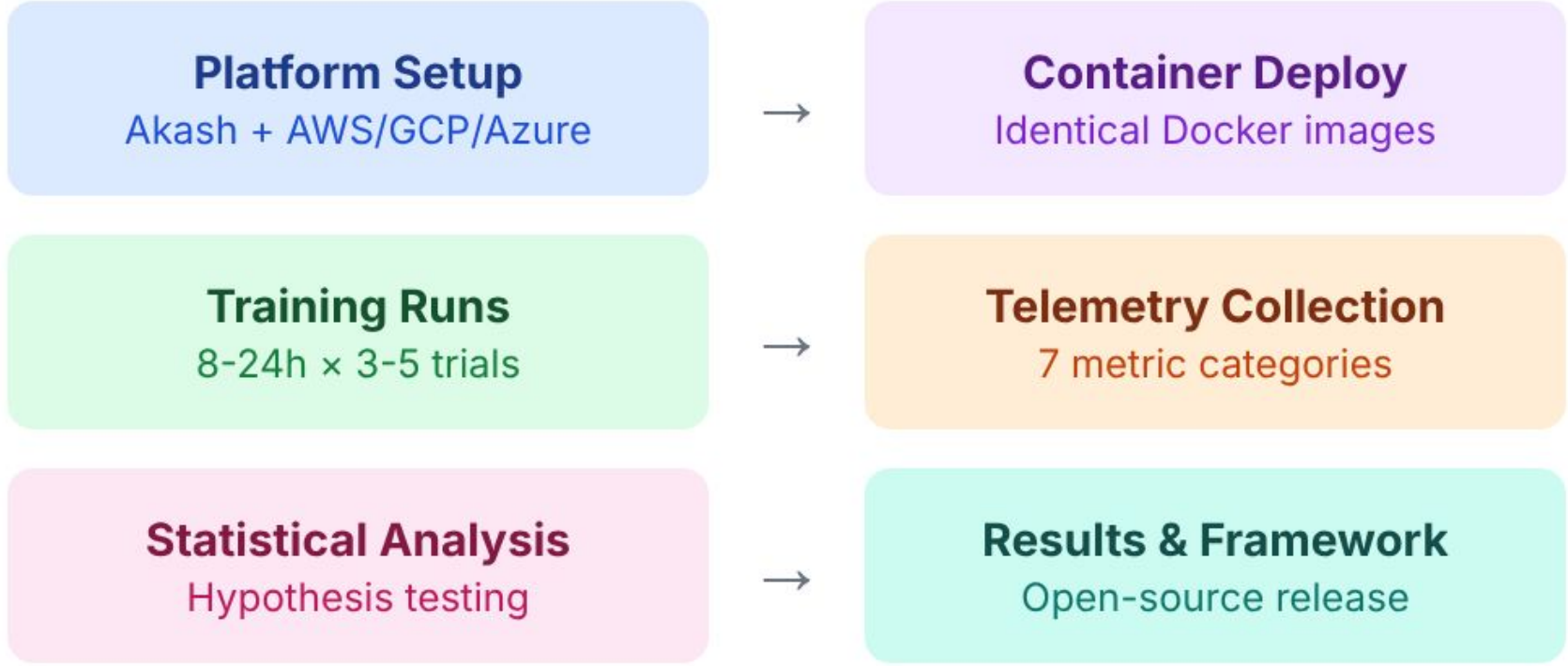


Fig. 2. Experimental Flow

VI. Conclusion

This research examines how decentralized GPU marketplaces stack up against traditional cloud providers for AI and machine learning training. It uses the same hardware, the NVIDIA H100, and runs consistent workloads, such as LLaMA-2 7B fine-tuning and vision training. The study will conduct benchmarks lasting 8 to 24 hours to measure throughput, latency, cost, reliability, and network performance. It addresses significant gaps by providing long-duration performance data, direct comparisons between marketplaces and cloud providers, and clear cost-per-work metrics. The results will highlight real bottlenecks, hidden costs, and benefits of each platform. Additionally, it will create a reusable benchmarking framework. In the end, this research will provide the first solid evidence to help researchers and organizations make informed choices about infrastructure for large-scale AI training.

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